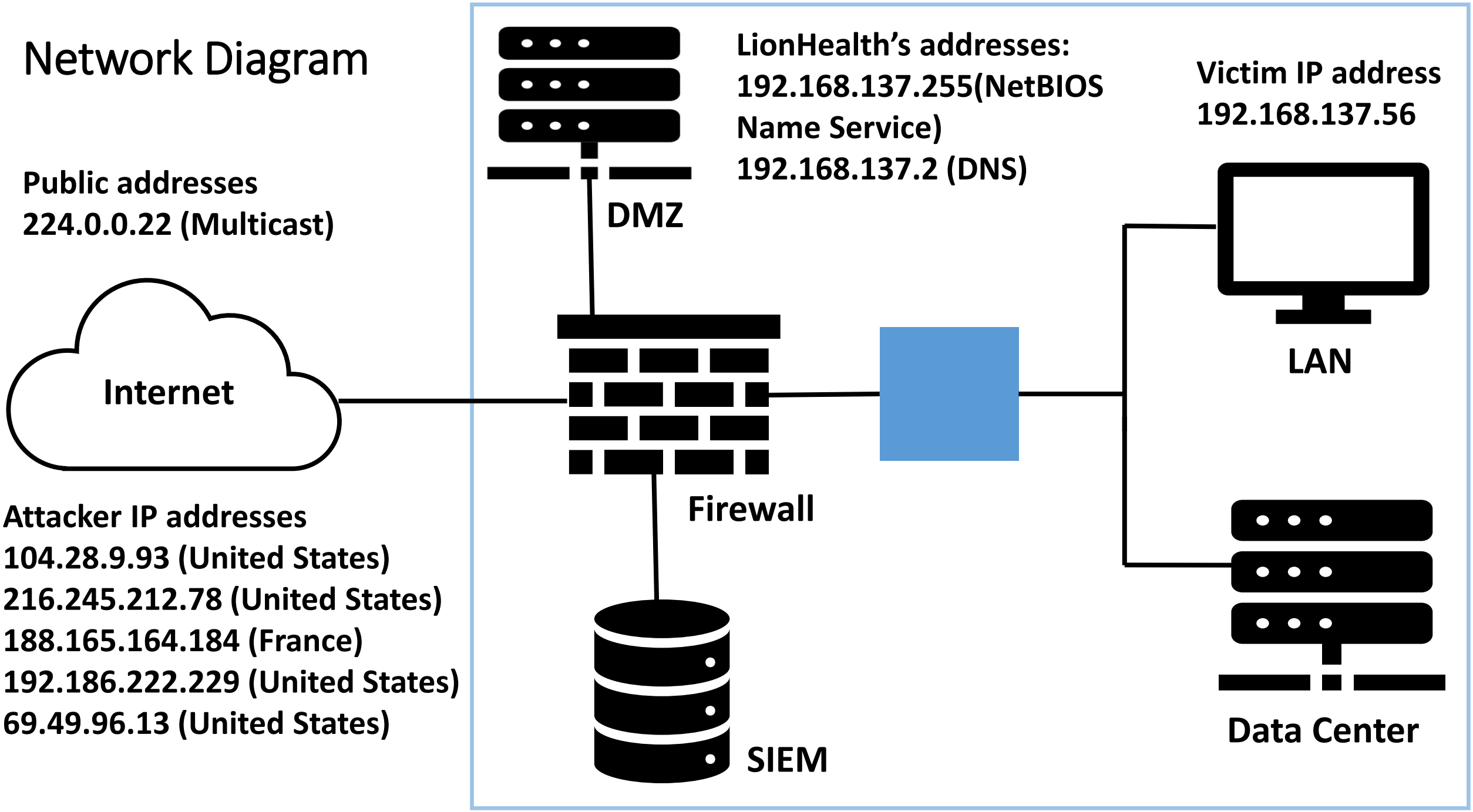
Digital Forensics - Security Operations Center (SOC) Incident Investigation Slides

Introduction

On September 11, 2015, LionHealth experienced a 4 minute and 44 second, high-severity cyberattack involving trojan malware, ransomware, and outdated software vulnerabilities



Network Diagram



Threats Identified

S/No.	Threat / Vulnerabilities found	Comments
1	Trojan/JS.Iframe.p	Trojan-activity
2	JS.Redirector.a	Trojan-activity
3	Trojan (CryptoWall)	Trojan-activity
4	Outdated Flash Version M1	Vulnerabilties

Mitigation Methods for Trojan/JS.Iframe.p

Description:

- A script that injects malicious iframe code into web pages to redirect users to harmful URLs or execute additional payloads. Targeting e.g. Compromised websites.

Removal:

- **Clear Temporary Files:** E.g. Disk Cleanup or CCleaner deletes all temporary internet and cache files where Trojan might reside.

- **Disable Suspicious Extensions:** Remove unknown or suspicious browse extensions, especially those added recently.

Prevention of Reinfection:

- **Regularly update:**

1. Website CMS and plugins.

2. Web server software.

3. Web browsers and extensions**.**

- **Disable Inline JavaScript:** Inline JavaScript is often exploited.

- **Disable iFrames:** prevent XSS attacks.

Mitigation Methods for JS.Redirector.a

Description:

- A malicious script injected into websites to redirect users to malicious or phishing sites.

Removal:

- **Run a Full System Scan:** Use antivirus e.g. Malwarebytes or Norton to perform a complete system scan.
- **Clear Browser Cache and History:** Malware can browser cache or history.
- **Reset Browser:** Ensures that any malicious redirects, extensions, or scripts are removed.

Prevention of Reinfection:

- **Keep Browsers and Extensions Updated:** Outdated browsers and extensions are often exploited by malware.
- **Use Browser Security Extensions:** Adblockers (e.g. uBlock Origin), Script Blockers (e.g. NoScript) prevent harmful scripts like this from running on untrusted websites.

Mitigation Methods for CryptoWall

Description:

A type of ransomware.

Removal:

- **Isolate the Infected Machine:** Disconnect machine from the network to prevent further damage.
- **Identify the Ransomware:** Look for a ransom note (e.g., DECRYPT_INSTRUCTION.html or DECRYPT_YOUR_FILES.txt) to find payment details.
- **Do Not Pay the Ransom:** Avoid paying as it doesn't guarantee file recovery and funds criminal activity. Focus on backups and identifying the attack source.
- **Perform a Full System Restore:** Restore from clean backups taken before the infection.

Prevention of Reinfection:

- **Use Anti-Ransomware Tools:** E.g. Malwarebytes Anti-Ransomware designed to detect ransomware behaviour like file encryption, process manipulation, and suspicious network activity.
- **Regular Backups:** Ensures availability from ransomwares like this.

Mitigation Methods for Outdated Flash Version M1

Description:

- Allows attackers to crash the victim's system, execute arbitrary code, install malware, control the system and steal confidential data.

Actions to take:

- Uninstall and disable Flash entirely
- Use alternatives to Flash like Lunascape which is (Free for macOS, Windows, iOS, and Android)

Conclusion

To prevent and combat the threats stated above, combine proactive measures (updates, security tools, backups) with reactive steps (system cleaning, threat isolation).

